

Session 6 — The Moon arrives

Practical Astronomy Intensive

6 questions 15 minutes 4 concepts 4 diagnostic

Terminator over full moon, impacts over volcanoes, 59% over 50% — plus the two things they should be able to DO: drive the scope to the terminator and bring back a focused phone frame of it.

1 A visitor asks when to bring their new telescope for the best view of lunar craters. What do you tell them?

lunar-observing

Recall

Multiple choice

- a At full moon — the whole disc is lit, so you see the most
REVEALS Full moon believed to be the best time — it is the flattest and worst
- b Any phase, but aimed at the bright middle of the disc where the light is strongest
REVEALS Craters hunted in the bright disc centre instead of at the terminator
- c** Around first quarter or a crescent, and look along the terminator, where low sunlight throws long shadows off every crater rim ✓
- d At full moon, with a Moon filter fitted; the filter is what brings the craters out
REVEALS Moon filter credited with revealing detail — it only tames glare

Full moon is sunlight straight down the line of sight: no shadows, no relief, a bland white glare. The terminator is where the Sun is rising or setting on the lunar surface, and a 1 km crater rim throws a shadow you cannot miss.

2 Mare Tranquillitatis is a large dark patch, smoother than the bright surrounding highlands. What is it?

lunar-geology

Apply

Multiple choice

- a A young surface, smooth because lunar weather has worn the craters away
REVEALS Moon's surface expected to erode like Earth's — no air, no water, nothing to erase a crater
- b A sea, or the dry bed of one that has since evaporated
REVEALS Maria taken as seas or former water — the 17th-century name read literally
- c** An ancient basalt lava plain flooding an old impact basin — dark and smooth because the lava resurfaced it ✓
- d A field of collapsed volcanic calderas, which is also what the craters are
REVEALS Craters believed volcanic rather than impact scars

Riccioli's 17th-century names stuck long after anyone believed in lunar seas. The maria are basalt; the craters around them are impact scars, which is why they come out round regardless of the impactor's approach angle — and why they are still there billions of years on.

- 3 The Moon is tidally locked, keeping one face towards Earth. Over many months, how much of its surface can we actually see from Earth, and why?

lunar-libration

Reason

Multiple choice

- a About 59% — libration rocks the Moon side to side and nods it up and down, letting us peek round each limb in turn ✓
- b Exactly 50% — tidal locking means the same half faces us and the rest never does
REVEALS Exactly half believed visible; libration ignored
- c 50%, since the other 41% is the permanently dark side that never receives sunlight
REVEALS Far side called the 'dark side' — it gets as much sunlight as the near side
- d It varies unpredictably, because the Moon wobbles randomly on its axis
REVEALS Libration pictured as a random wobble rather than predictable orbital geometry

Libration comes from an elliptical orbit (the Moon's spin is steady, its orbital speed is not) plus a tilted axis. It is predictable geometry — a libration chart tells you which limb is tipped toward you tonight, which is how anyone ever glimpses Mare Orientale. And the unseen 41% is the far side, not the dark side.

- 4 A student has a good phone and tonight's crescent Moon in the eyepiece. Which is the honest account of what that phone can and cannot do?

phone-astrophotography

Apply

Multiple choice

- a Zoom the phone in far enough and it will pull M31 out of the sky the same way it gets the Moon
REVEALS Phone zoom believed to reach faint deep-sky targets
- b Held at the eyepiece on a bracket it will give a genuinely good Moon shot on the first try; with night mode on a tripod it records constellations and, from a dark site, the Milky Way — but it will not reach a galaxy without a telescope in front of it ✓
- c A phone cannot do real astrophotography; it is a toy until you buy a DSLR
REVEALS Phone dismissed as incapable of real astrophotography
- d It will work at the eyepiece if you hold it steady enough; a bracket is for people with shaky hands
REVEALS Steady hand thought sufficient — the lens must sit at the exit pupil, which is what the bracket does

The phone's limits are real but not where people think. Night mode stacks several seconds and genuinely records sky. The eyepiece problem is not shake but geometry: the phone lens has to sit at the exit pupil, and that is a bracket's job, not patience.

- 5 The Moon is up and the scope is yours. Find the terminator, put a crater group on it in the eyepiece, and sketch what you see in five minutes — then tell the instructor why you chose that spot on the disc.

lunar-observing

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Watch that the student drives the scope to the terminator rather than the bright limb or the disc centre, centres a crater group where shadows are visibly long, and sketches shadow direction rather than outlines. The verbal check matters as much as the sketch: they should say the terminator is where the Sun is low on the lunar surface, and should NOT credit a Moon filter with revealing the detail. Points for noticing the shadows shorten over the session as the Sun climbs on that ground.

A pencil forces a student to actually look. Anyone who has sketched shadows along the terminator has stopped believing full moon is the good night.

- 6 Fit the phone adapter to the eyepiece and bring back one focused shot of the crater terminator — sharp enough that individual crater shadows are visible.

phone-astrophotography

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Watch for: adapter clamped to the eyepiece and the phone lens brought onto the exit pupil (the frame goes from a dark tunnel with a bright circle to a full field — the student should recognise that transition and adjust rather than shove the phone closer); exposure or brightness pulled DOWN, since the Moon blows out on auto; tap-to-focus on the terminator, not the bright limb; a shot taken at the terminator with visible shadow relief. Reject a hand-held blur or a white featureless disc — send them back to re-seat the bracket and drop the exposure.

The Moon is the one target where a phone at the eyepiece gives a real photograph on the first attempt. Everyone should go home with this frame — it is the hook for the whole imaging session on the last night.